Feeding Frenzy: testing mutual feeding benefits in dolphin–gannet hunting

Reuben Stanley | University of Western Australia

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1 Introduction and current evidence

Abstract
Dolphins and Cape gannets feed on the same fish schools, but their association does not establish a feeding advantage for both species. We test mutual feeding benefit in a three-dimensional agent-based model. Dolphins repeat local herding, a feasible strike and recovery. Development comparisons favoured an exposed-prey targeting rule for its response to adding gannets. Keeping that rule unchanged, fresh comparisons increased dolphin captures from 87 to 101 across four paired repetitions from two new schools. Matching gannet-only controls gave 26 captures versus 32 with dolphins. Both species gained in two of the four repetitions; gannet gains were concentrated in one repetition. A separate initial-depth contrast showed no monotonic reduction in gannet feeding: prey moved into and out of the reachable depth range. The pilot supports conditional joint gains under the specified rules, rather than a uniform reciprocal advantage. Capture calibration, numerical convergence and controlled attack timing remain open.

Motivation
The South African sardine run brings schooling fish into contact with several predators. Environmental conditions help explain the migration and aggregation of prey [1], while observations describe associations between feeding dolphins and gannets [2]. At the scale of one already-located school, the relevant question is whether those associations improve feeding returns, rather than merely bringing predators to the same resource.

Two possible services
Dolphins may concentrate fish and affect their vertical accessibility to diving birds. Gannet attacks may in turn disturb schooling and expose fish to dolphins. The second pathway is a hypothesis to test. Both predators also consume prey, and time spent herding or waiting can reduce feeding. Combined catch alone is therefore insufficient evidence that both species benefit.

Prior evidence
Models of successive predator attacks show how disturbance changes aggregations [3]. Underwater observations associate recent attacks with increased gannet feeding success [4], but do not supply the matched dolphin-only counterfactual needed here. The present contribution is a species-specific comparison of feeding returns under controlled initial conditions, not a claim that multiple-predator simulation is new.

Hypothesis and objectives
The overarching hypothesis is that dolphin–gannet hunting provides mutual feeding benefits: both species achieve greater individual feeding returns together than independently. The programme retains three objectives: (1) measure gannet feeding across initial prey depths; (2) establish dolphin herding and targeting rules, then test the effect of adding gannets; and (3) test whether controlled relative attack timing improves both returns. This draft reports a development and replication test for objective 2, followed by matching gannet-only controls and an initial-depth pilot for objective 1. Objective 3 remains a separate intervention.

2 Model and behavioural assumptions

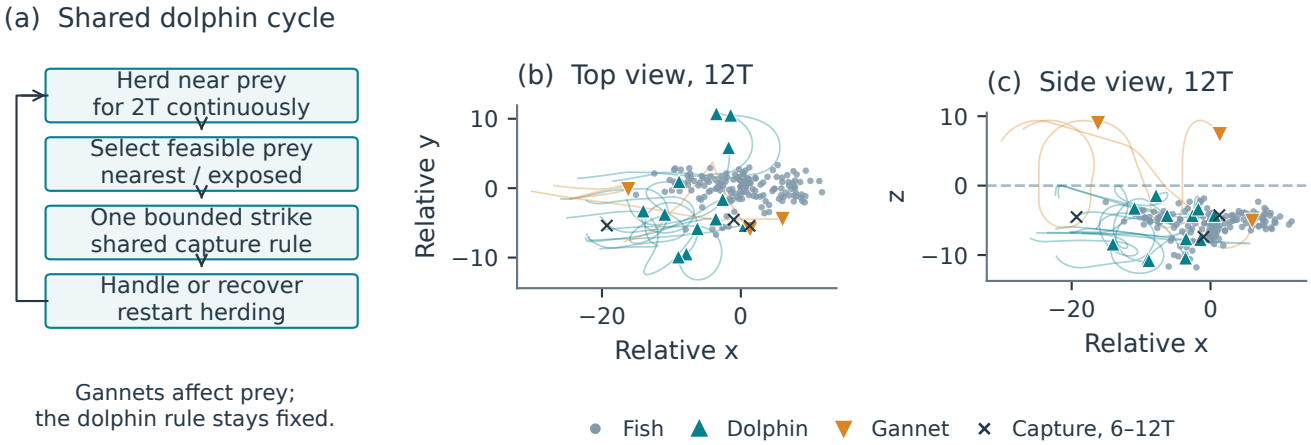


Figure 1: Model rules and a recorded encounter. (a) Dolphins repeat a near-prey herding period, one strike and recovery; birds affect dolphins through fish dynamics. (b,c) Recorded top and side views at 12T for development school 2 with exposed targeting and gannets. Paths cover 6–12T; crosses mark three actual captures. Coordinates are relative to the surviving prey centroid; z=0 is the surface. The snapshot illustrates behaviour, not verified whole-school containment.

Fish and domain

The model follows 160 point-represented fish in three dimensions. Horizontal motion is open and has a preferred migration direction. The surface is z=0, with water at negative z. Fish use a Couzin-type separation, alignment and attraction framework [5], extended with migration, habitat and predator-response terms. Repulsion has priority over social alignment and attraction. Fish speed is fixed at 3 model length units per T and native movement noise is zero. Nearby submerged predators alter escape steering, alignment, attraction and turning. Airborne birds provide no threat cue.

Dolphin herding cycle

Each of 12 dolphins follows an inherited local waypoint ahead of, beside and below a tracked fish patch. A feeding cycle requires 2T continuously within 4.5 units of at least one currently observed fish. Losing proximity resets the clock. Patch handoff or fragmentation does not itself reset it. This proximity criterion is a proxy for contributing pressure: it does not demonstrate a completed bait ball, complementary pod coverage or pressure on the same fish throughout.

Target selection

After the herding interval, a dolphin selects a physically feasible fish within range 6. The nearest rule ranks candidates by distance. The exposed rule first prefers fewer neighbours within radius 2, then lower predicted interception and turning time. Dense prey remain eligible. Each attack is bounded by a 6T bout, speed 6 and turning limit $\pi/2$ per T. Recovery lasts 2T after capture or 0.75T after failure, followed by a new herding cycle. Dolphins have exact fish positions, velocities and identities within range 25; they receive no bird-entry signal.

Gannet rules

Three gannets have global current prey information within detection depth 12. DBSCAN identifies groups using radius 2 and a minimum neighbourhood count of four. Eligible clusters are sampled in proportion to accessible fish count; the bird targets a dense accessible location within the selected group. Birds retain separate approach, plunge, submerged pursuit, ascent and recovery stages. Their physical dive and travel limits remain distinct from visibility. Observed Cape gannet dive trajectories motivate finite constraints [6], but do not calibrate model units or physiological maxima.

Capture

Food requires an eligible swept contact between point prey and an offset mouth sphere (radius 0.4, forward offset 0.3), followed by a fixed 0.7 draw. The first eligible contact closes the bout, successful or not; each captured fish is removed once and may differ from the selected target. Predators observe the same step-start fish state and propose motion. Fish advance under submerged-predator cues; contacts are resolved jointly in time order, then controller memory commits the actual resolved step. No density-dependent capture bonus is imposed.

3 Experimental design and development results

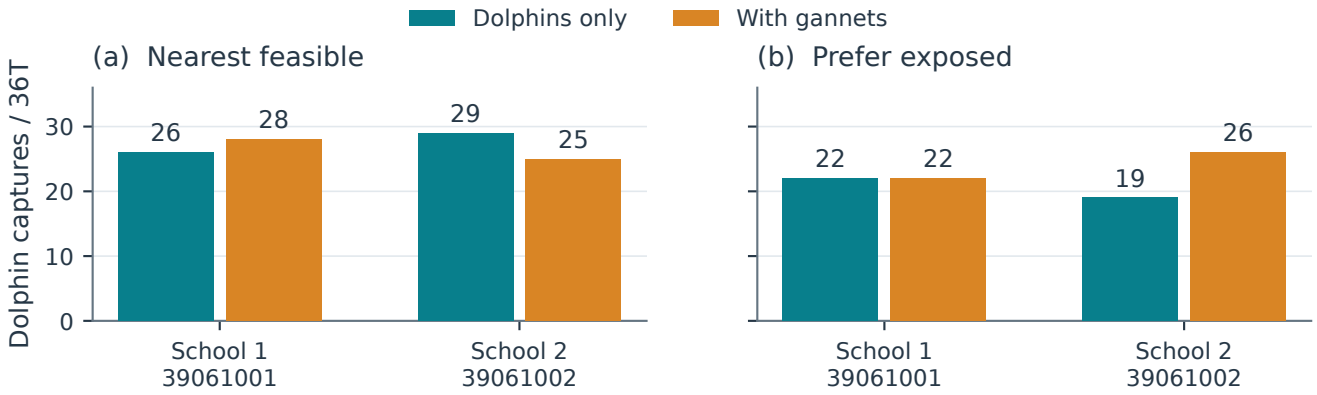


Figure 2: Development comparisons. Bars show dolphin captures in matched 36T encounters with 12 dolphins alone or with three gannets. Each target rule was tested on two development schools using one set of random draws. Nearest targeting produced totals of 55 alone and 53 with birds; exposed targeting produced 41 and 48. The initial response to adding gannets led us to select exposed targeting for follow-up. Individual outcomes are shown without confidence intervals.

Matched comparisons

Each policy was evaluated with dolphins alone (D) and with dolphins plus gannets (DG). Fish state, dolphin abundance, capabilities, initial readiness and controller state were matched within each pair. Paired encounters also began with matching random sequences for each dolphin's capture success and for resolving simultaneous contacts. The dolphin rule was identical between conditions. Gannets changed the fish state and competed for prey; they did not directly unlock dolphin attacks.

Outcome and duration

For species s, the feeding return is Y_s=C_s/(N_s T), where C_s is the number of uniquely removed fish credited to that species, N_s is its abundance and T is the complete encounter duration. The partner effects are Δ_D=Y_D(DG)-Y_D(D) and Δ_G=Y_G(DG)-Y_G(G). Joint benefit requires both to be positive in the same comparison. Each encounter lasts 36T, including travel, herding, unsuccessful attacks, handling and recovery. It starts from a predator-free school checkpoint after 20T of maturation; that preceding fish-only preparation is not dolphin herding time and is excluded from the feeding horizon.

Development set

The development comparison used two starting schools, labelled schools 1 and 2, and one set of random draws. Testing both target rules on both schools, with and without gannets, gave eight full encounters. The timestep was 0.00125T. No changes were made to the capture, fish-response or gannet rules between target policies. Requested herding is a fixed minimum cycle duration; it is not a prescribed fraction of total encounter time.

Observed effects

Nearest targeting changed dolphin captures from 26 to 28 in the first school and from 29 to 25 in the second. Exposed targeting changed 22 to 22 and 19 to 26. Its pooled gain was seven fish, or 17.1%, but it was not positive in both schools. Nearest targeting still produced more dolphin food in the mixed encounters, 53 versus 48. A larger response to partner presence is therefore distinct from maximum total feeding.

Selection rule and uncertainty

We selected exposed targeting for its response to adding gannets, although nearest targeting produced more captures with dolphins alone. Exposed mixed hunting also fell below nearest targeting alone on the development schools, 48 versus 55 captures. The follow-up fixes the exposed rule and tests its same-policy partner effect. It does not test whether mixed hunting beats the best-tested independent policy: nearest targeting was not evaluated on the fresh schools. With two development schools, the observed differences do not establish an optimal policy. Repeated runs from one school are not independent schools.

4 Replication on new schools

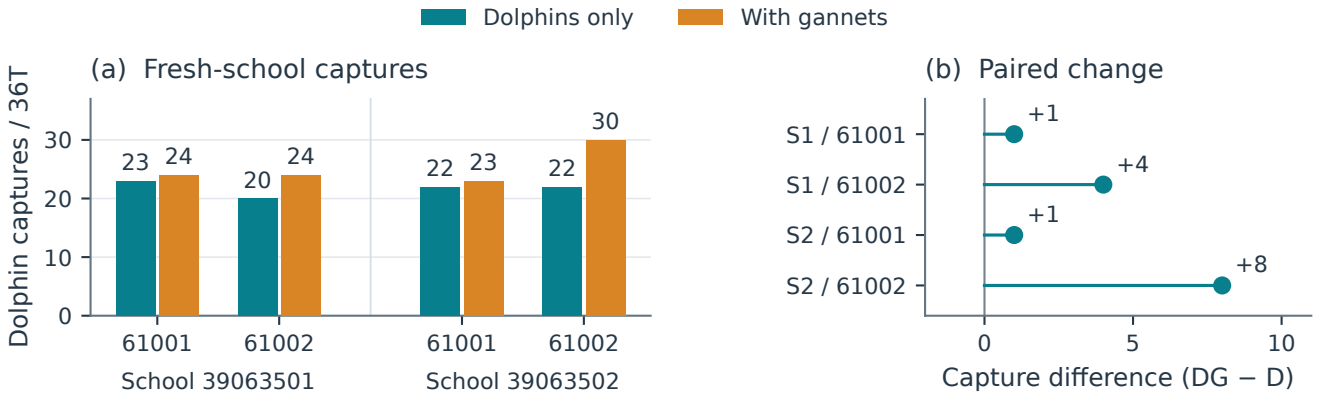


Figure 3: Replication of the unchanged exposed-prey rule on two new schools. (a) Bars show dolphin captures with and without gannets for two repetitions per school. (b) Points show the capture difference within each matched pair. The four paired comparisons come from two independent schools. The development data in Figure 2 are excluded from these replication estimates.

Follow-up design

We kept the dolphin and gannet rules unchanged for the follow-up tests. Two new schools, labelled new schools 1 and 2, were chosen before their feeding outcomes were known. Both used the original fish sampler and baseline scenario. At timestep 0.00125T, a checkpoint was saved after 20T without predators, followed by the original predator-free stability screen through 56T. Both retained all 160 fish and passed the finite, submerged, route-speed and connectivity criteria. No candidates were replaced.

Repeated comparisons

Each new school was tested in two repetitions, labelled repetitions 1 and 2. The repetitions used different random sequences for capture success, contact ties and each bird's target choice. Initial positions, velocities and readiness were unchanged between repetitions. Within each repetition, encounters with and without gannets began with the same dolphin rule and capture sequences. Paths and the number of random draws could then diverge as each encounter progressed.

Replication outcome

The follow-up tests produced 87 captures alone and 101 with gannets, a descriptive increase of 16.1%. The paired counts were 23 to 24 and 20 to 24 in new school 1, and 22 to 23 and 22 to 30 in new school 2. School-average gains were 2.5 and 4.5 fish. All four repetitions were positive, but they represent only two independent new schools.

Unit of replication

The eight encounters provide four paired presence comparisons but only two newly generated school configurations. Results are therefore presented for each school and repetition, with school-level means across the two repetitions. A pooled total is descriptive and is not a sample of four independent ecological schools. The earlier development gain is kept separate because it was used to select the target rule.

Verification

We checked each completed encounter for correct time budgets, unique prey removals, eligible mouth contacts, recovery periods and consistency with the saved source code. Each dolphin attack was checked against the event log to confirm at least 2T of continuous proximity to prey before launch. For exposed targeting, we checked that the selected fish had the fewest neighbours among feasible targets. We also replayed the saved random sequences for capture success, contact ties and bird target choice. These checks verify the implementation; they do not establish biological calibration or timestep convergence.

5 Gannet controls, depth and joint gains

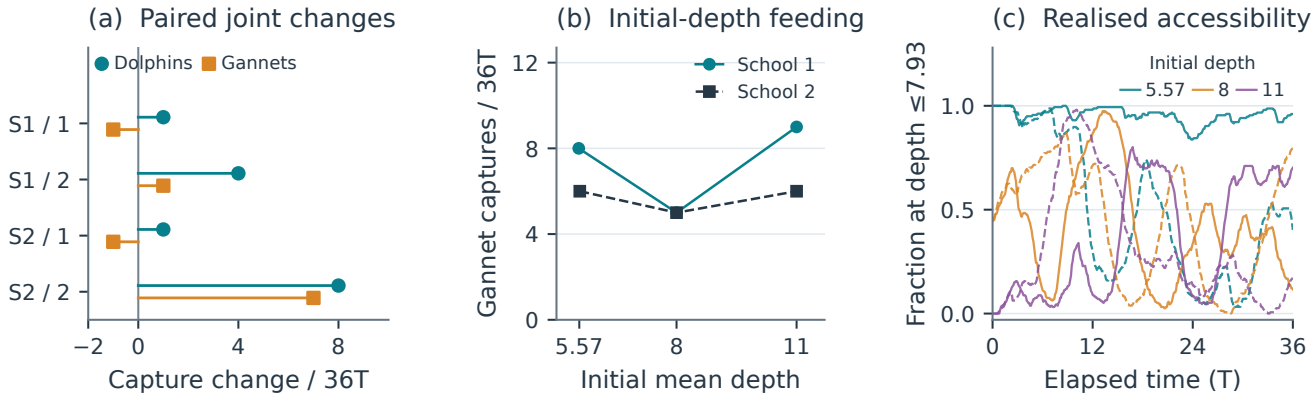


Figure 4: Matching gannet-only controls and initial-depth results. (a) Changes in dolphin and gannet captures when hunting together; S1 and S2 identify the two fresh schools, and 1 and 2 identify random settings 61001 and 61002. Both species gain in two of four repetitions. (b) Gannet-only captures across initial mean depths, using setting 61001; lines join the same school. (c) The fraction of surviving fish within the reference depth envelope changes during each encounter. Colour identifies initial depth; solid and dashed curves identify schools 1 and 2. Initial depth is not maintained, and the connecting lines in (b) are not fitted response curves.

Matching gannet controls

We completed eight additional 36T gannet-only encounters. Four unshifted controls match the two fresh schools and both random settings used in Figure 3. The three birds retain the same capabilities, initial states and private capture and target-choice streams as their mixed counterparts. Four further encounters translate each school to initial mean depths 8 and 11, using setting 61001. Translation preserves school geometry without clipping; the preferred fish depth remains 5 with deadband 3–7.

Joint feeding response

Gannet capture changes in the four unshifted comparisons were -1, +1, -1 and +7 fish. Both species gained in the second repetition of each school; in the other two, dolphin gains accompanied one fewer gannet capture. Pooled gannet captures increased from 26 to 32 (+23.1%), but the largest gain accounts for seven fish while the other three differences sum to -1. Mean gannet changes across repetitions were zero and +3 for the two schools. The matching dolphin means were +2.5 and +4.5. These outcomes show conditional joint gains, not a uniform reciprocal advantage.

Initial-depth result

The unshifted schools began at mean depths 5.56 and 5.58. At setting 61001, gannet captures were 8 and 6 without translation, 5 and 5 at initial mean depth 8, and 9 and 6 at initial mean depth 11. There was no monotonic decline. Initially, all unshifted fish, about 44–45% at mean depth 8, and none at mean depth 11 lay within the reference target-depth envelope of 7.93. Native prey movement then changed accessibility; the deepest starting treatment became reachable.

What depth does and does not test

Initial translation is a transient intervention. The fish retain their original habitat preference and can change depth throughout the encounter; selective consumption also changes survivor statistics. The depth envelope is a coarse geometric eligibility reference, not guaranteed interception or a calibrated physiological maximum. These runs therefore cannot establish a maintained-depth barrier or an optimal feeding depth.

Objective 3: attack timing

An isolated timing hook reproduced the disabled baseline exactly. The first enabled 8T check found no cue at any of six legal bird launches. Retrospective screening then identified one later opportunity in a different random setting. A separately declared, post-hoc 36T engineering run delayed one bird launch by 0.076T before loss of locality cancelled the wait. Its realised launch lag was 0.873T, short of the requested 1T. Dolphin and gannet captures, 24 and 11, matched the baseline. This demonstrates a partial deferral, without establishing a timing advantage.

6 Discussion, limitations and next steps

Interpretation
The development comparison shows that the sign of a partner effect depends on dolphin target choice and starting school. Exposed targeting produced a positive pooled response, whereas nearest targeting did not, although nearest targeting yielded more dolphin food overall. Fresh replication retained a positive dolphin response in both school averages and all four repetitions. The new gannet controls narrow the interpretation: both predators gained in two repetitions, but gannet returns did not improve in the other two. The central hypothesis remains falsifiable; mutual benefit may be restricted to particular prey states and behaviours.

Behavioural assumptions
The main uncertainty is whether 2T near fish represents useful herding effort and whether exposed prey are a reasonable targeting preference. The model does not assign complementary drivers and barriers, reserve targets or coordinate feeding turns. Herding represents contributions to within-pod collaboration through fish dynamics, while each individual uses the same local rule. Neither an attractive trajectory nor a labelled herding state proves sustained whole-school containment.

Perception and capture
Dolphins have exact local prey information, while birds have idealised global information within depth limits. The mouth geometry is simplified, prey are point-represented, and predator body-spacing avoidance is disabled. Capture success after eligible contact remains 0.7 for both species and is uncalibrated. The gannet field values of 13% and 36% describe success per observed attack technique [4], not conditional mouth-contact success and not first versus second attempts. They cannot be inserted directly as the model parameter.

Scope of the evidence
Time and length have no validated joint physical scale. The study omits learning, fatigue, hunger, satiety and energetic costs, and its finite prey supply is not replenished. Capture return is therefore not energetic profitability or fitness. Work on non-coordinating predators [7] and the broader seabird–subsurface-predator literature [8] also cautions against equating a feeding association with deliberate communication or reciprocal cooperation. A model gain identifies a mechanism worth testing, not evidence that real animals use the implemented rule.

Conclusion
The pilot establishes a reproducible comparison of specified herd–strike rules, with a positive dolphin response on two fresh schools and conditional gains for both predators. Gannet gains were uneven, and translating initial prey depth did not produce a monotonic feeding response because depth remained free to change. The next tests are a controlled timing intervention, more independent starting schools and sensitivity to the herding, perception and capture assumptions. These are needed before drawing a general ecological conclusion.

Reproducibility and draft status
The timestep is 0.00125T; a 36T encounter contains 28,800 steps. Configurations, source and checkpoint hashes, saved random states, event logs, trajectories and verified summaries are retained. The local Jupyter notebook verifies 24 sealed development, replication and gannet-only/depth encounters and regenerates all four report figures and eight tables. All 18 default code cells passed in fresh processes, including a relocated folder; analysis imports no native simulator. Original inputs and manifests are preserved. The unchanged optional runner passed a short simulation pair on the tested Linux host; platform constraints are documented. Timing checks have separate supporting evidence. The longer eight-school study is running and remains excluded from this snapshot. Code and evidence: https://github.com/reuben-trade/FishSim-Dolphin-Gannet-Pilot.

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